

DATA ANALYTICS LEVEL III

Day 10: Data Cleaning in SQL — Text Functions, Duplicates & Standardization

Student Activities

How to Use This Document

The dirty customer list from Day 9 has been found INSIDE the company database (of course it has — the CSV was exported from here). Power Query can't fix the source; you can. Connect SQLTools to Day10_practice.db (download from the LMS). For every question, record BOTH the SQL and the result. The Surgeon's Rules apply to everything you do today.

Activity	What You Submit	Time
Activity 1: Diagnose (SELECT only)	Queries + findings for the 6 checks	45 min
Activity 2: The Surgery	Your full cleaning script (.sql) + before/after counts	70 min
Activity 3: Prove & Report	Queries + answers for the 7 report items	45 min
Learning Journal Entry #9	Your journal reflection	20 min

Activity 1 — Diagnose (SELECT Only — Change Nothing)

Goal: Build the complete dirt report using read-only queries. If you type UPDATE in this activity, you have already failed it.

1. Duplicates: compare COUNT(*) with COUNT(DISTINCT customer_id). How many duplicate rows hide in the table?
2. Invisible spaces: using WHERE col <> TRIM(col), find every row with space-dirt in full_name, city, or email. (Show the values wrapped in brackets: '[' || city || ']'). How many rows total?
3. Spelling chaos: SELECT DISTINCT city — how many raw variants exist, and what should the clean list be?
4. Missing contacts: who has NULL email? (Names + IDs.)
5. The lie: compute SUM(CAST(monthly_spend AS INTEGER)) on the dirty table. Write the number down. Then compute what CAST('1,500' AS INTEGER) returns. Explain in one sentence why the total is so wrong.
6. Fill the Dirt Report table (your findings, one line per dirt type — this becomes your cleaning plan).

Dirt Type	Evidence Query	Rows Affected	Planned Fix
Duplicates			
Spaces			
Casing/variants			
NULL emails			
Text-numbers			

Submit: Day10_Activity1_[YourName].docx

Activity 2 — The Surgery

Goal: Clean the table in place, by the Rules, keeping a script of everything you ran.

Step 0 — Rule 1: Backup

7. CREATE TABLE customers_backup AS SELECT * FROM customers; — then verify it has 28 rows.

Step 1 — Spaces and casing

8. One UPDATE to TRIM full_name, city, and email on every row. (Why is no WHERE acceptable HERE? Answer in a comment: trimming a clean value changes nothing.)
9. Lowercase every email with UPDATE + LOWER.

Step 2 — Standardize the cities (Rule 2 every time!)

10. For EACH city: first SELECT the rows your WHERE will catch, then UPDATE. Use the LOWER(city) trick so one statement catches all casings. Cities: Manila, Quezon City (include 'qc!'), Cebu, Davao, Makati.
11. Verify: SELECT DISTINCT city — exactly 5 clean names remain?

Step 3 — Free the numbers

12. Remove the commas from monthly_spend with UPDATE + REPLACE.
13. Verify: SUM(CAST(monthly_spend AS INTEGER)) now. It is much bigger than the Activity 1 “lie” — but it is STILL not the final truth. Write it down and answer: why is it not yet equal to Day 9's Power Query total? (Hint: what have you not removed yet?)

Step 4 — Deduplicate by creation

14. CREATE TABLE customers_clean AS SELECT DISTINCT * FROM customers; — then COUNT it (expected: fewer than 28) AND re-run SUM(CAST(monthly_spend AS INTEGER)) on customers_clean.

NOW does it match Day 9's Power Query total? Explain in one line what the duplicates were doing to your earlier total.

15. Rule 4 check: COUNT(customers) vs COUNT(customers_backup) vs COUNT(customers_clean) — write all three and explain each number.

Submit: Day10_Surgery_[YourName].sql — every statement in order, with -- comments, including your SELECT previews

Activity 3 — Prove & Report

Goal: Show the cleaned data answers business questions correctly — and matches Day 9's Excel result.

16. The cross-tool proof: unique customers, distinct cities, and total spend from customers_clean. Do all three match your Power Query results from Day 9? (They must: same truth, two workbenches.)
17. Customers per city: one COUNT query per city (5 queries, Day 6 style). Which cities tie for the most customers?
18. Follow-up list: using IFNULL, produce full_name + contact where missing emails show as 'NO EMAIL — follow up'. Who appears?
19. Spend tiers: write ONE SELECT with CASE that labels every customer High (≥ 3000), Mid (1500–2999), or Low (< 1500), showing name, spend, tier.
20. Count each tier (three COUNT queries with the same CAST conditions). High + Mid + Low should equal your customer total.
21. Top 3 spenders from the clean table (ORDER BY the CAST value, DESC, LIMIT 3).
22. CHALLENGE — the handover note: in 3–4 sentences, write what you would tell the next analyst: what was dirty, what you changed, where the backup is, and which numbers they can now trust.

Submit: Day10_Activity3_[YourName].docx (or .sql with comments)

Learning Journal Entry #9

Goal: Reflect on cleaning with power tools.

- 1. State the four Surgeon's Rules from memory, and which one exists because of the “no WHERE” horror story.**
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- 2. Explain the CAST trap:** why did the dirty total say ₱4,542, and what two-function combination fixed it?
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3. Power Query cleaning vs SQL cleaning: name one advantage of each. When would you choose which?

4. My confidence with SQL cleaning today (1 = lost, 5 = very confident), and why:

Submit as: Day10_Journal9_[YourName].docx

Submission Checklist

- ☐ **Activity 1:** six diagnostic findings + the Dirt Report table
- ☐ **Activity 2:** the full surgery script with SELECT previews and comments
- ☐ **Activity 3:** seven report answers incl. the handover note
- ☐ Learning Journal Entry #9